A Heuristic Approach for Optimum Cost and Layout Design of 3D Reinforced Concrete Frames

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Abstract

This paper presents a new methodology for cost optimization of the preliminary layout design of three-dimensional reinforced concrete (RC) frames. This approach is capable of being easily employed for the optimal layout design of a realistic large RC structure that accounts for constraints imposed by design standards. The new approach considers modeling, structural analysis, concrete member design, and discrete optimization together with data on cost of systems and materials. The methodology comprises two parts. First, using the cross-sectional action effects as design variables, a heuristic cost function is presented as an alternative to traditional cost functions for layout optimization of RC structures. Using the presented cost function, a structural optimization problem is formulated for column layout design of 3D RC frames. Then, an Ant System (AS) algorithm, a discrete method, is proposed to solve the cost optimization problem. Two comparative design examples are included to demonstrate the performance of the new methodology and the excellent convergence of the Ant Colony Optimization (ACO) algorithm.

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Introduction

In the design of framed buildings, achieving an economical column layout is one of the primary objectives. In fact, preliminary layout design of a frame impacts on the entire design process and consequently, the total costs of the building. In order to select the most economical plan among various layouts that satisfies architectural requirements, one needs to be aware of the reciprocal effects of cost factors and layout variables. That is, achieving an economical layout plan, particularly in reinforced concrete structures, requires simultaneous consideration of cost and layout variables.

Cost optimization of RC structures has, over the past decades, been developed using a variety of methods. However, one limiting feature of existing methods is that the representation is limited to structures with predefined shapes. A review of past work done in the field of optimal design of RC structures shows that a large fraction of the early works are mostly confined to cost optimization of individual members (Sarma and Adeli 1998).

A number of studies have been published on either cost optimization or layout optimization of RC structures. Balling and Yao (1997) and Zou et al. (2007) considered the optimization of the RC frames. Shaw et al. (2008) developed a method for columns layout for orthogonal buildings. Nimtawat and Nanakron (2009; 2010) proposed a genetic algorithm based method for beam–slab layout design of rectilinear floors without considering the cost elements. Zhu and Zhang (2010) presented a new method for designing the structures layout. Sahab et al. (2005a; 2005b) considered the cost and topological optimization of flat slab buildings. There are also a significant number of published studies on the cost optimization of RC beams and columns (Sarma and Adeli 1998). Even so, the reciprocal effects of cost and topology of large RC structures have not been adequately considered. Cost optimization of structures with predefined shapes may lead to sub-optimal results as they suffer from the problem of not necessarily having an

optimum starting point, and layout optimization of structures without considering all involved cost factors may not lead to globally optimal results either.

In this study, a methodology is proposed to deal with the cost optimization problem of column layout design of 3D RC frame buildings employing an ACO algorithm. The first ACO algorithm was the Ant System (AS) developed by Dorigo et al. (2004) which has successfully been applied to many combinatorial optimization problems. Camp et al. (2005) proposed an Ant based algorithm for the design of steel frames. Kaveh et al. (2008) presented an ACO algorithm for structural topology optimization.

The Problem Definition

Depending on the nature of the optimization problem, the process of achieving an optimum solution might be much quicker, shifting from one design space to another by changing design variables as the space dimensions. That is, one may shift from one set of design variables to another that causes a new definition for the objective function and may lead to changes in constraints and even vary their nature from design variables to behavioral ones and vice versa.

In concrete structures, at least three different cost factors should be considered in optimization: costs of concrete, steel reinforcement, and the formwork. The effects of these factors on the optimal cost can be imposed on the weighted coefficients of the cost function. So, the general cost function for a reinforced concrete member can be expressed in the following form:

$$C = c_c A_c + c_{sl} A_{sl} + c_{sv} A_{sv} + c_f P_f \tag{1}$$

Where c_c, c_{sl}, c_{sv} and c_f are the unit costs of concrete, longitudinal steel, shear steel and formwork respectively and A_c, A_{sl}, A_{sv} and P_f are their corresponding quantities. In a 3D frame consisting of beams and columns Eq. (1) can be written as follows:

$$\begin{cases} C^{(b)} = c_c A_c^{(b)} + c_{sl} A_{sl}^{(b)} + c_{sv} A_{sv}^{(b)} + c_f P_f^{(b)} & (2.1) \\ C^{(c)} = c_c A_c^{(c)} + c_{sl} A_{sl}^{(c)} + c_{sv} A_{sv}^{(c)} + c_f P_f^{(c)} & (2.2) \\ C = \sum_{1}^{N_b} C^{(b)} + \sum_{1}^{N_c} C^{(c)} & (2.3) \end{cases} \quad (2)$$

where the superscript (c) stands for the columns parameters and the superscript (b) for beams and N_c and N_b are the number of columns and beams in the frame respectively.

In most published studies, the cost optimization functions deal only with cross-sectional variables. In layout design of structures, the cross-sectional variables are functions of design action effects which are not determinate and vary as the shape changes. Therefore, in an iterative procedure to solve an optimization problem, each step includes dealing with both the structural analysis and structural design variables. In such cases, unless alternative design variables are selected for the cost function, the optimization procedure might be too unwieldy.

This situation holds as long as parameters such as cross-sectional variables are mainly obtained from implicit functions of structural analysis outputs based on the suggested relations and constraints in the design standards. On the other hand, design standards do not uniquely provide the exact values for these cross-sectional parameters and they are not obtained from an explicit mathematical procedure.

In a layout optimization problem, in addition to the classic cross-sectional variables the layout of a structure would be variables of the problem. If the reciprocal relationships between the cross-sectional design factors and the design action effects are determined, the cost can be represented as a function of design action effects. In the design process, having the design action effects in critical sections, all the cross-sectional parameters and consequently, the total cost based on Eq. (3) can be calculated. To put it simply, using structural analysis outputs, say internal actions of a

member, as design variables has advantages over using structural design outcomes such as cross-sectional characteristics of a beam. Firstly, design action effects of each section can be easily obtained from structural analysis, and in an iterative mathematical procedure, re-analyzing a structure is considerably less time-consuming and more precise than re-designing the structure. Moreover, using action effects, the cost function will be considered in a section rather than a member. It enables the designer to select a number of sections for each member and in the whole structure to control the cost, and there is no necessity to conduct the optimization process over the entire member.

Therefore the aim is to explore the relation between the variations of action effects of RC members with the variations of the cross-sectional parameters and determine how these two types of variables affect each other. Then, based on such relations, a new cost function is derived that is a function of action effects rather than cross-sectional parameters. In order to formulate the structural optimization problem, this study makes use of the requirements specified in the Australian standard for concrete structures AS3600 (2009), which is based on the limit state design method of concrete structures. Formulation for frames is conducted on two groups of structural elements separately: beams and columns.

Mathematical Formulation for Beams

For a beam to be designed, three action effects are taken into consideration: positive bending moment, negative bending moment and shear forces. Consider an arbitrary section of a rectangular reinforced concrete beam as shown in Fig. 1. The dimensions of the section are b and $h^{(b)}$, the areas of tension and compression reinforcement are $A_{st}^{(b)}$, $A_{sc}^{(b)}$ and the area of shear reinforcement in a unit length of beam is $A_{sv}^{(b)} / s$.

Now, consider Eq. (3) as a potential alternative cost function to Eq. (2.1) in an arbitrary RC beam section.

$$C_i^{(b)} = c_1 M_{u_i}^{+(b)} + c_2 M_{u_i}^{-(b)} + c_3 V_{u_i}^{(b)} \tag{3}$$

The capacity or the ultimate strength of the section in negative and positive flexure and shear for section i are $M_{u_i}^{-(b)}$,$M_{u_i}^{+(b)}$ and $V_{u_i}^{(b)}$ respectively, which can be obtained from Eq. (4) through Eq. (6) in a balanced section in which both tension and compression steel yield.

$$M_u^{+(b)} \cong A_{sc}^{(b)} f_{yl} \left(d^{(b)} - d_c^{(b)}\right) + (A_{st}^{(b)} - A_{sc}^{(b)}) f_{yl} \left(d^{(b)} - a/2\right) \tag{4}$$

$$\begin{cases} M_u^{-(b)} \cong A_{sc}^{(b)} f_{yl} (d^{(b)} - d_c^{(b)}) \\ d_c^{(b)} = 0.5\gamma k_u d^{(b)} \end{cases} \tag{5}$$

$$V_u^{(b)} \cong \frac{f_{yv} d A_{sv}^{(b)}}{s} + \beta b d^{(b)} (f_c^{'})^{0.5} \tag{6}$$

where f_{yl} is the yield strength of the longitudinal reinforcement, f_{yv} is the yield strength of the shear reinforcement, f'_c is the characteristic compressive cylinder strength of concrete at 28 days, $d_c^{(b)}$ is the distance from the extreme compression fiber of the concrete to the compressive force. The coefficients γ and k_u are calculated based on the characteristic strength of the concrete and reinforcing steel, β is a coefficient based on the standard, and s is the centre-to-centre spacing of shear reinforcement. Other parameters are shown in Fig. 1.

Design standards impose some upper and lower bounds for the amount of longitudinal and shear reinforcement, and consequently the value of bearing capacities that need to be considered. In this study the doubly reinforced condition, as an inclusive case, is considered. In case of singly reinforcement the current methodology could be easily employed by considering $A_{sc}^{(b)}$=0.

In order to shift from Eq. (2-1) to Eq. (3) and come up with the set of $\{c_1, c_2, c_3\}$, the first step is to determine how cross-sectional variations of $A_c^{(b)}, A_{sl}^{(b)}, A_{sv}^{(b)}$ and $P_f^{(b)}$ affect the section bearing capacities $M_u^{+(b)}$, $M_u^{-(b)}$ and $V_u^{(b)}$, and vice versa. That is, the reciprocal relationships between these two sets of variables need to be determined, and find out how varying the amount of each cross-sectional parameter influences the section bearing capacities and how one should change the cross-sectional parameters to vary section capacities. Given the unit costs c_c, c_{sl}, c_{sv} and c_f, the cost function can be defined using Eq. (2.1) for each section of a beam. The parameters $A_c^{(b)}, A_{sl}^{(b)}, A_{sv}^{(b)}$ and $P_f^{(b)}$ are obtained from the sum of the cross-sectional variables for all beams. If any of cross-sectional parameters changes, the cost function varies as follows

$$\Delta C^{(b)} = c_c \Delta A_c^{(b)} + c_{sl}\, \Delta A_{sl}^{(b)} + c_{sv}\, \Delta A_{sv}^{(b)} + c_f\, \Delta P_f^{(b)} \tag{7}$$

On the other hand, using Eq. (8) variations in section capacities would change the cost function as follows

$$\Delta C^{(b)} = c_1 \Delta M_{u_i}^{+(b)} + c_2\, \Delta M_{u_i}^{-(b)} + c_3\, \Delta V_{u_i}^{(b)} \tag{8}$$

Eqs. (7) and (8) show the contribution of each factor to cost changes and sensitivity of the cost to each term. For example, changing a unit of $A_c^{(b)}$ causes a change of c_c units in cost. Therefore, if the effect of variations of $A_c^{(b)}, A_{sl}^{(b)}, A_{sv}^{(b)}$ and $P_f^{(b)}$ on the variations of $M_u^{+(b)}$, $M_u^{-(b)}$ and $V_u^{(b)}$ are determined, the contribution of each section capacity to cost changes, that is the set of $\{c_1, c_2, c_3\}$, can be determined.

To simplify the calculations, two assumptions are made. First, since we intend the section to remain in a balanced condition, any changes in the amount of compression steel reinforcement are considered equal to those of tension steel reinforcement. That is:

$$\Delta A_{st}^{(b)} = \Delta A_{sc}^{(b)} = \Delta A_{sl}^{(b)} \tag{9}$$

The second assumption is that the depth of the compressive region of concrete, the variable a, is not significantly affected by variations of the section width b. The width of a beam section is generally constrained by a number of factors including architectural constraints, standards limitations and limitations of columns and joints. All these constraints cause the width of the section to have a minor variation during the optimization process. It does not mean that the variation of width is ignored, but in practice, such a variation is so limited that it has a negligible effect on the depth of the compressive region. Moreover, since the cover to steel reinforcement is considered to be constant, the variations of section height equal the variations of $d^{(b)}$.

According to the calculations presented in the Appendix, the set of $\{c_1, c_2, c_3\}$ in Eqs. (3) and (8) can be presented with regard to the set of predefined cost coefficients $\{c_c, c_{sl}, c_{sv}, c_f\}$ in Eq. (10).

$$\begin{cases} c_1 = \frac{1}{3}c_c K_4 + \frac{1}{2}c_{sl}K_1 + \frac{2}{3}c_f K_7 \\ c_2 = \frac{1}{3}c_c K_3 + \frac{1}{2}c_{sl}K_1 + \frac{2}{3}c_f K_6 \\ c_3 = \frac{1}{3}c_c K_5 + c_{sv}K_2 + \frac{2}{3}c_f K_8 \end{cases} \tag{10}$$

The obtained coefficients c_1, c_2 and c_3 determine how the variations of bearing capacities $M_{u_i}^{+(b)}$, $M_{u_i}^{-(b)}$ and $V_{u_i}^{(b)}$ contribute to the variation of cost function. K_1 trough K_8 are calculated based on the equations presented in the Appendix.

Mathematical Formulation for Columns

Consider an arbitrary section of a square reinforced concrete column as shown in Fig. 2. The dimensions of the section are $h^{(c)}$, the area of the longitudinal reinforcement is $A_{sl}^{(c)}$ and the area of shear reinforcement in a unit length of column is $A_{sv}^{(c)} / s$.

Now, consider Eq. (11) as a potential alternative cost functions to Eq. (2.2) in an arbitrary RC column section.

$$C_i^{(c)} = c_4 N_{u_i}^{(c)} + c_5 M_{u_i}^{(c)} + c_6 V_{u_i}^{(c)} \tag{11}$$

where $N_{u_i}^{(c)}$, $M_{u_i}^{(c)}$ and $V_{u_i}^{(c)}$ are the axial load-carrying capacity, bending moment capacity and the shear capacity of the *i*th column section respectively. If an appropriate set of $\{c_4, c_5, c_6\}$ could be found, in such a way that Eq. (11) represents the cost of the section, the design variables would shift from $A_c^{(c)}$, $A_{sl}^{(c)}$, $A_{sv}^{(c)}$ and $P_f^{(c)}$ to $N_u^{(c)}$, $M_u^{(c)}$ and $V_u^{(c)}$.

The capacity or the ultimate strength of the section in axial compression, flexure and shear for a column section can be obtained from Eq. (12) through Eq. (14).

$$N_u^{(c)} \cong 0.85 \gamma k_u f_c^{'} h^{(c)} d^{(c)} + E_s \sum_i^n A_{sl_i}^{(c)} \varepsilon_i \tag{12}$$

$$M_u^{(c)} = N_u^{(c)} e \cong 0.85 \gamma k_u f_c^{'} h^{(c)} d^{(c)} e_c + E_s \sum_i^n A_{sl_i}^{(c)} \varepsilon_i e_i \tag{13}$$

$$V_u^{(b)} \cong \frac{f_{yv} d^{(c)} A_{sv}^{(c)}}{s} + \beta h^{(c)^2} (f_c^{'})^{0.5} \tag{14}$$

in which E_s is the modulus of elasticity of reinforcement, $A_{sl_i}^{(c)}$ and e_i are respectively the cross-sectional area of reinforcement and the distance from the extreme compression fiber for each bar, n is the number of bars in a column, ε_i is the strain in reinforcement. The parameters e and e_c

are the eccentricity of the axial force from the centroidal axis and the distance from the extreme compression fiber for compressive force in the concrete of the cross-section, respectively. Other parameters are shown in Fig. 2. Obviously, if the column section is subject to a biaxial bending moment or shearing forces bending moment and shear force . a considered for both axes.

In order to shift from Eq. (2-2) to Eq. (11) and come up with the set of $\{c_4, c_5, c_6\}$, the reciprocal relationships between the cross-sectional parameters, $A_c^{(c)}, A_{sl}^{(c)}, A_{sv}^{(c)}, P_f^{(c)}$, and axial force capacity $N_u^{(c)}$, bending moment capacity $M_u^{(c)}$ and shear force capacity $V_u^{(c)}$, should be determined, as for beams in Section 2.1.

Given the unit costs c_c, c_{sl}, c_{sv} and c_f, the cost function can be defined using Eq. (2.2) for the entire column. The parameters $A_c^{(c)}, A_{sl}^{(c)}, A_{sv}^{(c)}$ and $P_f^{(c)}$ are obtained from the sum of the cross-sectional variables for all columns. If any of cross-sectional parameters changes, the cost function varies as follows

$$\Delta C^{(c)} = c_c \Delta A_c^{(c)} + c_{sl}\, \Delta A_{sl}^{(c)} + c_{sv}\, \Delta A_{sv}^{(c)} + c_f\, \Delta P_f^{(c)} \tag{15}$$

On the other hand, using Eq. (11) variations in section capacities would change the cost function as follows

$$\Delta C^{(c)} = c_4 \Delta N_u^{(c)} + c_5 \Delta M_u^{(c)} + c_6 \Delta V_u^{(c)} \tag{16}$$

Eqs. (15) and (16) show the contribution of each factor to cost changes and sensitivity of the cost to each term. Therefore, if the effect of variations of $A_c^{(c)}, A_{sl}^{(c)}, A_{sv}^{(c)}$ and $P_f^{(c)}$ on the variations of $N_u^{(c)}$, $M_u^{(c)}$ and $V_u^{(c)}$ are determined, the contribution of each section capacity to cost changes, that is the set of $\{c_4, c_5, c_6\}$, can be found.

In order to simplify the calculations, some assumptions are made. The first one is that any changes in cross-sectional area of longitudinal reinforcement bars are the same for all bars in a column. That is, in each column for all the bars in that column:

$$\Delta A_{sl_i}^{(c)} = \Delta A_{sl_j}^{(c)} \tag{17}$$

According to Eq. (17), in order to simplify the mathematical formulation and determine the relationship between the variations of cross-sectional parameters and bearing capacities, the changes in all column bars are considered the same.

The second assumption is that if the column section is not square, any changes in the length of the cross-section of columns equal the changes in the width. When changing columns dimensions in such a way, the second degree variations are neglected. Besides, since the cover to steel reinforcement is considered to be constant, the variations of section dimensions equal the variations of $d^{(c)}$. The above-mentioned assumptions are defined in Eqs. (18) and (19).

$$\Delta h_1^{(c)} = \Delta h_2^{(c)} = \Delta h^{(c)} = \Delta d^{(c)} \tag{18}$$

$$\Delta A_c^{(c)} = \Delta(h^{(c)^2}) = 2h^{(c)}\Delta h^{(c)} \tag{19}$$

According to the calculations presented in the Appendix, the set of $\{c_4, c_5, c_6\}$ in Eqs. (11) and (16) can be presented with regard to the set of predefined cost coefficients $\{c_c, c_{sl}, c_{sv}, c_f\}$ in Eq. (20).

$$\begin{cases} c_4 = \frac{1}{3}c_c K_{12} + \frac{1}{2}c_{sl} K_9 + \frac{1}{3}c_f K_{15} \\ c_5 = \frac{1}{3}c_c K_{13} + \frac{1}{2}c_{sl} K_{10} + \frac{2}{3}c_f K_{16} \\ c_6 = \frac{1}{3}c_c K_{14} + c_{sv} K_{11} + \frac{2}{3}c_f K_{17} \end{cases} \tag{20}$$

The obtained coefficients c_4, c_5 and c_6 determine how the variations of bearing capacities $N_{u_i}^{(c)}$, $M_{u_i}^{(c)}$ and $V_{u_i}^{(c)}$ contribute to the variation of cost function. K_9 trough K_{17} are calculated based on the equations presented in the Appendix.

Formulation of Optimization Problem for Frames

Having the relevant coefficients to optimize a frame, one can use Eqs. (3) and (11) instead of Eq. (2.3). In this case, the cost is the sum of the cost functions of all selected sections in the structure based on Eqs. (3) and (11). Using these function as alternative cost functions to Eq. (2.3), the design variables can be shifted from cross-sectional variables to action effect ones. For the structural optimization problem, the strength constraints on each selected section *i*, whether for the beams or the columns, under a load case may be written as

$$\begin{cases} \varnothing N_{u_i} \geq N_i^* \\ \varnothing M_{u_i} \geq M_i^* \\ \varnothing V_{ui} \geq \left|V_i^*\right| \end{cases} \tag{21}$$

in which, N_i^* , M_i^* and $\left|V_i^*\right|$ are the axial force, the flexure and shear action effects of section *i*, either in columns or beams, and $\varnothing$ is the strength reduction factor. The serviceability requirements limit the maximum deflection u_{max} on the entire member to Δ_{max} under the serviceability load case. So, for all sections:

$$u_{max\,i} \leq \Delta_{max\,i} \tag{22}$$

Other constraints for durability, fire resistance, minimum cover and minimum flexural strength, can be easily added to the problem, based on relevant design codes.

For a 3D RC frame under an arbitrary loading system f to be optimally designed for cost and layout, the general formulation for the structural optimization problem can be written as follows:

$$\begin{cases} \underset{l_1,l_2,\ldots,l_{NSP}}{min} \quad Cost(l_1,l_2,\ldots,l_{NSP}) = \sum_{1}^{NBS} C_i^{(b)} + \sum_{1}^{NCS} C_i^{(c)} \\ \text{----------} \\ s.t \begin{cases} strength\, constraints\, based\, on\, Eq.(21) \\ serviceability\, requirements\, based\, on\, Eq.(22) \\ \{l_{min}\} \le \{l_i\} \le \{l_{max}\} : for\, all\, spans \\ other\, constraints\, based\, on\, standards \end{cases} \end{cases} \tag{23}$$

in which NBS and NCS are the total number of control beams and columns sections in the structure respectively, $C_i^{(b)}$ and $C_i^{(c)}$ are obtained from Eqs. (3) and (11) and l_{min} and l_{max} are the maximum and minimum lengths of each span, which might be applied in order to satisfy architectural considerations, and NSP is the total number of spans in x and y directions.

There are several ways to deal with the above cost optimization problem. Although optimizing the layout of a frame is a continuous problem, it can be dealt with as a discrete problem by discretizing the domain. In fact, in the design process, we usually deal with the dimensions as discrete sizes, and in practical problems, it is often the case that design variables must be chosen within a discrete manufacturer's inventory. The dimensions of concrete sections are usually varied by a certain size, e.g. 25 mm or 50 mm a step, which makes the section dimensions discrete. Therefore, one can define the optimization problem of RC sections as a discrete optimization problem rather than a continuous one. In the next section, an Ant Colony Algorithm, as a discrete optimization tool, is proposed to solve the problem.

The Developed Ant Colony Algorithm

The aim of the developed ACO algorithm in this study is to find the optimum column layout locations of a 3D RC frame which leads to the optimum spans of the frame in both the x and y directions, as shown in Fig. 3, in a way that optimizes the relevant costs. The height of stories, that is column lengths, is considered to be determinate and constant. Ant System (AS), as the first ACO algorithm, has successfully been applied to many combinatorial optimization problems during the past decade. Many of the present Ant Colony Optimization (ACO) algorithms are the extensions of the Ant System (AS) algorithm. Two main phases of the AS algorithm are the ants' solution construction and the pheromone update.

As the first step, the domain needs to be discretized. As such the construction graph is defined assuming that that span lengths are bounded in [L_{min} L_{max}] in either direction (L_{min} and L_{max} might be different between the x and y direction) with intervals (accuracy) equal to ε. That is, each L_i rests in the set of $\{L_{min}, L_{min} + \varepsilon, L_{min} + 2\varepsilon, \ldots, L_{max} - \varepsilon, L_{max}\}$. Obviously, the smaller ε is, the more accurate the results will be and the more running time the algorithm needs. As shown in Fig. 3, an L-shape graph is defined as the construction graph to demonstrate the plan of the 3D frame in x-y plane. The members on the horizontal and vertical wings of the graph denote spans in the x and y directions respectively. In order for the construction graph to stand for the structure layout, the total number of spans and the number of nodes need to be N_x+N_y and (N_x+N_y+1) respectively, in which N_x and N_y are the total number of spans in the x and y directions respectively. Hence, the construction graph is defined as a graph with N_x+N_y+1 nodes and $((L_{max} - L_{min})/\varepsilon * N_x)+((L_{max} - L_{min})/\varepsilon * N_y)$ edges. Each pair of nodes are connected with $(L_{max} - L_{min})/\varepsilon$ edges, i.e. there are $(L_{max} - L_{min})/\varepsilon$ choices for each ant on each node to select another one, for making its trail (solution). It should be pointed out that, as stated in the definition of constraints and shown in Fig. 3, L_{max} and L_{min} might be specified differently for each span. In this case, the

number of choices for ants on each node might differ from the other ones. Now that the construction graph is prepared, ants can start their solution process in the following phases (Kaveh and Sharafi 2008a).

On the construction graph the initial magnitude of pheromone must be set. In AS a good heuristic to initialize the pheromone trails is to set them to a value slightly higher than the expected amount of pheromone deposited by the ants in one iteration. A rough estimate of the value can be obtained by setting $\forall(i, j) \quad \tau_{ij} = \tau_0 = m / C^{nn}$, where m is the number of ants, and C^{nn} is the cost of a tour generated by the nearest neighbor heuristic (Kaveh and Sharafi 2008b).

Now, two groups of artificial ants are located on the first and last nodes of both wings of the construction graph, i.e. Nodes 1 and N_x+1on the horizontal wing and Nodes 1 and N_y+1 on the vertical wing to construct their solution. The lengths of paths are introduced, and the relevant parameters and the termination criteria are defined. Each value (i,j) of pheromone matrix and heuristic matrix shows the tendency of the ant located on Node i of the construction graph to choose the Edge j to move towards the next node. Since there are as many as $(L_{max} - L_{min})/ \varepsilon$ edges that connect every two successive nodes, the dimension of these two matrices will be $(N_y+N_x)*(L_{max} - L_{min})/ \varepsilon$.

For organizing the heuristic matrix and initial pheromone matrix, we can make use of structural analysis experience. As boundary conditions and constraints, plus any loads can affect the length of spans, the entries of heuristic matrix and initial pheromone matrix might be organized based on such parameters. Moreover, in order to save time, some criteria like the symmetry of the plan are considered when the heuristic matrix is defined. In this case, the entries of heuristic matrix relating to preferred choices might receive higher numbers relative to the entries of the other ones. Using such a heuristic matrix or defining the initial pheromone matrix using the above-

mentioned structural rules helps the algorithm converge sooner. Thus, each value (i,j) of choice information matrix, which is obtained by multiplying the corresponding arrays of heuristic matrix by those of pheromone matrix, shows the tendency or desirability of the ant located on Node i of the construction graph to choose Edge j to move towards Node $i+1$.

The next phase consists of two segments that form the basis of the heuristic. In the first segment of this phase, ants located on the starting nodes of the construction graph, whether Node 1 or Node $N+1$ (N stands for N_x on the horizontal wing and for N_y on the vertical one) on either wing, start to construct their solutions. At each construction step, Ant k applies a probabilistic action choice rule based on Eq. (24), known as random-proportional rule, to decide which edge to visit next.

$$p_{ij}^{k} = \begin{cases} \dfrac{\left[\tau_{ij}\right]^{\alpha}\left[\eta_{ij}\right]^{\beta}}{\sum_{l \in N_i^k}\left[\tau_{il}\right]^{\alpha}\left[\eta_{il}\right]^{\beta}} & \text{if } j \in N_i^k \\ 0 & \text{if } j \notin N_i^k \end{cases} \tag{24}$$

That is, the ant leaves its source Node $i \in \{1,2,\ldots,N+1\}$ on the construction graph to destination node on the connecting Edges $j \in \{1, 2, \ldots, (L_{max} - L_{min})/\varepsilon\}$, using the random proportional rule, similar to the random roulette wheel rule. The probability of Edge j to be chosen by an ant on Node i is proportional to the entry (i, j) of the choice information matrix. This relationship forms the basis of the AS algorithm and shows that if Ant k is positioned on Node i, it will move to the next Node j with the probability of ρ_{ij}^{k}. In this relationship, τ_{ij} is the magnitude of pheromone on the trails and η_{ij} is the heuristic value. α and β are two parameters which determine the relative influence of the pheromone trail and the heuristic information. N_i^k is the feasible neighborhood of Ant k when being at Node i, that is, the set of edges that Ant k is allowed to choose as its next destination, and is decided depending on the problem in hand. The probabilistic choice of the

next edge then works analogously to the roulette-wheel selection procedure of evolutionary computation. The quantity of pheromone trail and heuristic value on the edges determine the size of the slice on a circular roulette wheel. Obviously, the higher pheromone or higher heuristic value on the edges increases the possibility of selecting the corresponding node. The ants continue constructing their solution and move from a node to the next one, until Node $N+1$ on both wings as the end of their trail. The constraint for ants on the horizontal wing is that the sum of all spans must be equal to L_x, and for those on the vertical wing, L_y. Therefore, two groups of ants are located on both ends of both wings of the graph, so that every edge has the same overall probability of being chosen by the ants.

After the segment of random choice, the values of N_x+N_y lengths and thereby the new geometrical layout for the x-y plane is obtained. Based on the required accuracy for the calculations, a number of control sections are set for each member (for example, one at each member end and one at midspan) (Kaveh and Sharafi 2009). Through the structural analysis method for frames, and using the behavior constraints, design actions effects in the identified control sections are determined and using other constraints, the required bearing capacities for each control section are calculated. Based on the bearing capacity of control sections, the cost function or the cost of this solution is obtained.

The last phase consists of updating information and statistics, and then updating pheromone trails. Using the information obtained in the phase of constructing the solution. The trail presenting the best cost, up to this stage, is selected as the best so-far iteration. Then pheromone trails of construction graph are updated. This act is done by first lowering the pheromone value on all arcs (pheromone evaporation), and then adding pheromone on the edges the ants have

crossed (pheromone depositing). Pheromone evaporation is done by lowering the pheromone value on all edges by a constant factor ρ according to Eq. (25):

$$\tau_{ij} \leftarrow (1-\rho)\tau_{ij} \quad \forall (i,j) \in A \tag{25}$$

In this segment the parameter ρ depends on the number of steps defined for termination criteria, and initial pheromone and heuristic values. Then, depositing pheromone is done by increasing the pheromone value on all edges. For this purpose, ants are allowed to deposit pheromone on the edges associated with the best so-far iteration and the quality of the solutions in each step according to Eq. (26).

$$\tau_{ij} \leftarrow \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k} \ , \ \forall (i,j) \in A \tag{26}$$

where $\Delta\tau_{ij}^{k}$ is the pheromone added to τ_{ij} by Ant k, and defined as follows:

$$\Delta\tau_{ij}^{k} = \begin{cases} 1/C^{k} & \text{if edge } (i,j) \text{ belongs to } T^{k} \\ 0 & \text{otherwise} \end{cases} \tag{27}$$

The value of C^k is taken as the cost of the best trail or the best iteration and T^k is the set of suitable solutions. For depositing pheromone, the pheromone values on the edges associated with the best iteration up to now increase, and the Ant deposits an amount of pheromone equal to *1/cost* on the edges of construction graph associated with the best so-far iteration; where *cost* is the value of cost function obtained from best iteration.

The above procedure continues until the termination criterion is satisfied; then the optimum layout will be the best-so-far solution for spans. In each step, a basic structural analysis is required to calculate the section actions. Making use of a structural analysis approach, say the finite element method or the simplified methods based on design standards, such a calculation takes a small amount of time for each section. Using Eq. (2-3) as the objective function would be

significantly more time-consuming than using Eqs. (3) and (11). When Eq. (2-3) is used, due to the variation of lengths in each step, the estimation of design variables in each step would be considerably encumbered; apart from the fact that, in each step, the algorithm would iteratively need to deal with the design formulas. Eqs. (3) and (11) make the optimization process more practicable in the layout optimization of large structures.

Numerical Examples

To demonstrate the robustness of the proposed approach, two examples are presented. All computations were performed on P9700 @2.80 GHz computer running MATLAB R2009b. In order to ensure that the obtained solution is global or near global optimum, many runs were made in parallel. Since each run is fully independent of the others, the program can be run in parallel so that the total execution time will be practically the same as required for a single run.

Example 1: The column layout of a four-storey reinforced concrete frame building, each of 2.90 m height with a plan as shown in Fig. 4(a) is further optimized in the present work. The live load is 5.0 kN/m^2 and the dead load, excluding the self-weight of concrete, is 2.5 kN/m^2. The average unit price for concrete is assumed to be 55 units/m^3, and 3900 units/m^3 for reinforcing steel. The average unit price for formwork is 20 units/m^2. Other design parameters used in this example are the characteristic tensile strength of steel reinforcement f_y = 460 N/mm^2, the characteristic strength of concrete f'_c = 35 N/mm^2, and the cover of steel bars c=25 mm. The permissible spans are defined within the bounds of L_{max} = 7.5 m and L_{min} = 3.5 m. As a primary design for the proposed ACO algorithm, considered spans are $L_{x1} = L_{x2} = L_{x3} = L_{x4}$ = 5.0 m and $L_{y1} = L_{y2} = L_{y3} = L_{y4} = L_{y5}$ = 5.0 m, It should be added that every initial design based on preliminary judgment of the designer and/or using approximate charts or formulas, which meet the design standard

requirements, can be used as the initial design and as the starting point of optimization process. Based on the primary design, the cost of the frame structure computed using Eq. (2-3) is 26512 units.

For this example, three control sections are selected for every member. Using the above-obtained equations the values of K_1 to K_{18}, and consequently the values of c_1 to c_6 for all selected columns and beams sections are obtained. Having the necessary coefficients, the optimization problem can be formulated based on Eq. (23).

The Ant algorithm attempts to find the optimum spans based on the objective function and relevant constraints. After 105 iterations, and at CPU time of 33.8 seconds, the optimum lengths of L_{x1} = 4700 mm, L_{x2} = 5300 mm, L_{x3} = 5300 mm and L_{x4} = 4700 mm in the x direction, and L_{y1} = 5800 mm, L_{y2} = 6400 mm and L_{y3} = 5800 mm in the y direction are obtained. Having determined the optimum lengths, sections are designed based on AS3600 (2009) resulting in a total cost of 25239 units.

The present methodology thus provides a further cost saving of 4.8% over the previously optimized design of Sahab et al. (2005b). It should be noted that in Sahab's work, first, the optimization process is carried out for a predefined shape and then the optimum number of equal spans is obtained employing a genetic algorithm.

Example 2: Consider the plan of a three story building as shown in Fig. 4(b). This example was dealt with in a report on comparative costs of concrete framed buildings (Goodchild 1993) that has been recommended as a benchmark for future studies. The conventional design of this example has been carried out and optimized by a team of professional structural engineers. The total length and width of the building are 37.5 m and the heights of stories are 3.95 m. The live

load on intermediate floors is 5.0 kN/m^2 and on the roof is 1.5 kN/m^2. Dead loads are self-weight and the imposed dead load of 1.5 kN/m^2. The average unit price for concrete is assumed to be 53.5 units/m^3, and 3120 units/m^3 for reinforcing steel. The average unit price for formwork is 18.5 units/m^2. The permissible spans are defined within the bounds of L_{max} = 8.5 m and L_{min} = 5 m. Goodchild (1993) suggests the cost of 95833 units for $L_{x1} = L_{x2} = L_{x3} = L_{x4} = L_{x5}$ = 7.5 m and $L_{y1} = L_{y2} = L_{y3} = L_{y4} = L_{y5}$ = 7.5 m, based on BS8110 (1997). For this example, three control sections are selected for every member.

Considering the above design as the primary design for proposed ACO algorithm, the Ant algorithm attempts to find the optimum spans based on the objective function and observing other relevant constraints. After 120 iterations, and at CPU time of 49.3 seconds, the optimum lengths of $L_{x1} = L_{y1}$ = 7200 mm, $L_{x2} = L_{y2}$ = 7800 mm, $L_{x3} = L_{y3}$ = 7700 mm, $L_{x4} = L_{y4}$ = 7800 mm and $L_{x5} = L_{y5}$ = 7200 mm are obtained. Having determined the optimum lengths, sections are designed based on AS3600 (2009) resulting in a total cost of 92568 units.

For both examples, the termination criterion for the ACO algorithms is defined as the number of iterations, when the improvement in the solution quality was less than 0.02% after ten consecutive iterations. Figs. 5(a) and 5(b) show typical convergence histories of the proposed ACO algorithm for the examples.

Conclusion

This paper presents a new layout optimization model for the cost optimization of the column layout design of 3D reinforced concrete frames. The proposed approach simplifies the optimization process when the topology optimization problem considers the involved cost elements. The proposed algorithm for solving the cost optimization problem is an Ant System based algorithm, which can be easily employed for optimizing the span lengths and column

locations and is also capable of working with different cost functions. The presented examples show that the proposed algorithm using the new cost optimization function provides satisfactory results.

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Appendix

Mathematical Calculations for Eqs. (10) and (20)

1. Beams: Calculations for Eq.(10)

Any changes in the longitudinal steel $A_{sl}^{(b)}$ results in changes in the capacities of a beam section as follows:

$$\begin{cases} \dfrac{\Delta M_u^{-(b)}}{\Delta A_{sl}^{(b)}} \cong f_{yl}\left(d^{(b)} - d_c^{(b)}\right) \rightarrow \Delta A_{sl}^{(b)} = (f_{yl}\, d^{(b)}(1-0.5\gamma k_u))^{-1}\, \Delta M_u^{-(b)} = K_1\, \Delta M_u^{-(b)} \\ \dfrac{\Delta M_u^{+(b)}}{\Delta A_{sl}^{(b)}} \cong f_{yl}\left(d^{(b)} - d_c^{(b)}\right) \rightarrow \Delta A_{sl}^{(b)} = (f_{yl}\, d^{(b)}(1-0.5\gamma k_u))^{-1}\, \Delta M_u^{+(b)} = K_1\, \Delta M_u^{+(b)} \\ \dfrac{\Delta V_u^{(b)}}{\Delta A_{sl}^{(b)}} \cong 0 \rightarrow \Delta V_u^{(b)} \; and \; \Delta A_{sl} \; are\, independent\, of\; each\, other \end{cases} \tag{28}$$

If $A_{sv}^{(b)} / s$ changes the variation of the beams bearing capacities are

$$\begin{cases} \Delta M_u^{-(b)} \; and \; \dfrac{\Delta A_{sv}^{(b)}}{s} \; are\, independent\, of\; each\, other \\ \Delta M_u^{+(b)} \; and \; \dfrac{\Delta A_{sv}^{(b)}}{s} \; are\, independent\, of\; each\, other \\ \dfrac{\Delta V_u^{(b)}}{\dfrac{\Delta A_{sv}^{(b)}}{s}} \cong f_{yv} d \rightarrow \dfrac{\Delta A_{sv}^{(b)}}{s} \cong (f_{yv}\, d)^{-1}\, \Delta V_u^{(b)} = K_2\, \Delta V_u^{(b)} \end{cases} \tag{29}$$

If the effective area of the beam section $\Delta A_c^{(b)}$, i.e. *bd* varies:

$$\Delta A_c^{(b)} = \Delta(bd) = b\,\Delta d + d\,\Delta b + \Delta b\,\Delta d = b\,\Delta d \tag{30}$$

$\Delta M_u^{-(b)}$ is not a function of *b,* and based on the second assumption, the dependence of $\Delta M_u^{+(b)}$ on the changes of *b* is neglected. Moreover, the first term of $V_u^{(b)}$ is not dependent on *b* as well. So, the variations of the section capacities due to variation of cross-section area are

$$\frac{\Delta M_u^{(b)}}{\Delta A_c^{(b)}} = \frac{\Delta M_u^{(b)}}{\Delta(bd)} \cong \frac{\Delta M_u^{(b)}}{b(\Delta d)} \tag{31}$$

$$\begin{cases} \dfrac{\Delta M_u^{-(b)}}{\Delta(bd^{(b)})} \cong \dfrac{f_{yl} A_{sc}^{(b)}}{b}(1-0.5\gamma k_u) \rightarrow \Delta A_c^{(b)} = \Delta\left(bd^{(b)}\right) \cong \left[f_{yl} \dfrac{A_{sc}^{(b)}}{b}(1-0.5\gamma k_u) \right]^{-1} \Delta M_u^{-(b)} = K_3 \ \Delta M_u^{-(b)} \\ \dfrac{\Delta M_u^{+(b)}}{\Delta(bd^{(b)})} \cong \dfrac{f_{yl} A_{st}^{(b)}}{b}(1-0.5\gamma k_u) \rightarrow \Delta A_c^{(b)} = \Delta\left(bd^{(b)}\right) \cong \left[f_{yl} \dfrac{A_{st}^{(b)}}{b}(1-0.5\gamma k_u) \right]^{-1} \Delta M_u^{+(b)} = K_4 \ \Delta M_u^{+(b)} \\ \dfrac{\Delta V_u^{(b)}}{\Delta(bd^{(b)})} \cong \dfrac{f_{yv} A_{sv}^{(b)}}{bs} + \beta(f'_c)^{1/2} \rightarrow \Delta A_c^{(b)} = \Delta(bd^{(b)}) \cong \left[\dfrac{f_{yv} A_{sv}^{(b)}}{bs} + \beta\left(f'_c\right)^{0.5} \right]^{-1} \Delta V_u^{(b)} = K_5 \, \Delta V_u^{(b)} \end{cases}$$

(32)

The variation of the perimeter of a rectangular beam section, which determines the variation of the beam formwork, affects the section capacity as follows:

$$\Delta P_f^{(b)} = \Delta b + 2\Delta d^{(b)} \tag{33}$$

$$\begin{cases} \dfrac{\Delta M_u^{-(b)}}{\Delta b + 2\Delta d^{(b)}} \cong f_{yl} A_{sc}^{(b)} \rightarrow \Delta P_f^{(b)} = \Delta b + 2\Delta d^{(b)} \cong 2\left[f_{yl} A_{sc}^{(b)}(1-0.5\gamma k_u) \right]^{-1} \Delta M_u^{-} = K_6 \, \Delta M_u^{-} \\ \dfrac{\Delta M_u^{+(b)}}{\Delta b + 2\Delta d^{(b)}} \cong f_{yl} A_{st}^{(b)} \rightarrow \Delta P_f^{(b)} = \Delta b + 2\Delta d^{(b)} \cong 2\left[f_{yl} A_{st}^{(b)}(1-0.5\gamma k_u) \right]^{-1} \Delta M_u^{+} = K_7 \, \Delta M_u^{+} \\ \dfrac{\Delta V_u^{(b)}}{\Delta b + 2\Delta d^{(b)}} \cong \dfrac{1}{\dfrac{f_{yl} A_{sv}^{(b)}}{s} + \beta b(f'_c)^{1/2}} + \dfrac{1}{\beta d(f'_c)^{1/2}} \rightarrow \Delta P_f^{(b)} = \Delta b + 2\Delta d^{(b)} \cong [\dfrac{2}{\dfrac{f_{yv} A_{sv}^{(b)}}{s} + \beta b(f'_c)^{1/2}} + \dfrac{1}{\beta d(f'_c)^{1/2}}] \Delta V_u = K_8 \, \Delta V_u \end{cases}$$

(34)

Now, that the reciprocal relationships between the variations of cross-sectional variables and strength capacity parameters of a beam section are obtained, multiplying both sides of Eqs. (1.A) by $c_{sl}/2$, Eqs. (2.A) by c_{sv} , Eqs. (5.A) by $c_c/3$ and Eqs. (7.A) by $c_f/3$ and then adding them up result in:

$$c_c\,\Delta A_c^{(b)} + c_{sl}\,\Delta A_{sl}^{(b)} + c_{sv}\,A_{sv}^{(b)} + c_f\,\Delta P_f^{(b)} = \left(\frac{1}{3}c_c K_4 + \frac{1}{2}c_{sl}K_1 + \frac{2}{3}c_f K_7\right)\Delta M_u^{+(b)} + \left(\frac{1}{3}c_c K_3 + \frac{1}{2}c_{sl}K_1 + \frac{2}{3}c_f K_6\right)\Delta M_u^{-(b)} + \left(\frac{1}{3}c_c K_5 + c_{sv}K_2 + \frac{2}{3}c_f K_8\right)\Delta V_u^{(b)} \tag{35}$$

Comparing Eq. (35) with Eqs. (7) and (ACI-Committee-318, Institute et al.) results in Eq. (10):

2. Columns: Calculations for Eq. (20)

Any changes in the longitudinal steel $A_{sl}^{(c)}$ results in changes in the capacities of a column section as follows:

$$\begin{cases} \dfrac{\Delta N_u^{(c)}}{\Delta A_{sl}^{(c)}} \cong E_s \sum_i^n \varepsilon_i \rightarrow \Delta A_{sl}^{(c)} = (E_s \sum_i^n \varepsilon_i)^{-1}\,\Delta N_u^{(c)} = K_9\,\Delta N_u^{(c)} \\ \dfrac{\Delta M_u^{(C)}}{\Delta A_{sl}^{(c)}} \cong E_s \sum_i^n \varepsilon_i e_i \rightarrow \Delta A_{sl}^{(c)} = (E_s \sum_i^n \varepsilon_i e_i)^{-1}\,\Delta M_u^{(c)} = K_{10}\,\Delta M_u^{(c)} \\ \dfrac{\Delta V_u^{(c)}}{\Delta A_{sl}^{(c)}} \cong 0 \rightarrow \Delta V_u^{(c)}\ and\ \Delta A_{sl}^{(c)} are\, independent\, of\ each\, other \end{cases} \tag{36}$$

With respect to $A_{sv}^{(c)}/s$, the variations of the column capacities are

$$\begin{cases} \Delta N_u^{(c)}\ and\ \dfrac{\Delta A_{sv}^{(c)}}{s}\ are\, independent\, of\ each\, other \\ \Delta M_u^{(c)}\ and\ \dfrac{\Delta A_{sv}^{(c)}}{s}\ are\, independent\, of\ each\, other \\ \dfrac{\Delta V_u^{(c)}}{\dfrac{\Delta A_{sv}^{(c)}}{s}} \cong f_{yv} d^{(c)} \rightarrow \dfrac{\Delta A_{sv}^{(c)}}{s} \cong (f_{yv}\, d^{(c)})^{-1}\,\Delta V_u^{(b)} = K_{11}\,\Delta V_u^{(b)} \end{cases} \tag{37}$$

If the cross-section area of the column varies:

$$\begin{cases} \dfrac{\Delta N_u^{(c)}}{\Delta A_c^{(c)}} \cong 0.85\gamma k_u f'_c \rightarrow \Delta A_c^{(c)} \cong (0.85\gamma k_u f'_c)^{-1}\,\Delta N_u^{(c)} = K_{12}\,\Delta N_u^{(c)} \\ \dfrac{\Delta M_u^{(c)}}{\Delta A_c^{(c)}} \cong 0.85\gamma k_u f'_c e_c \rightarrow \Delta A_c^{(c)} \cong (0.85\gamma k_u f'_c e_c)^{-1}\,\Delta M_u^{(c)} = K_{13}\;\Delta M_u^{(c)} \\ \dfrac{\Delta V_u^{(c)}}{\Delta A_c^{(c)}} \cong \dfrac{f_{yv} A_{sv}^{(c)}}{2hs} + \beta (f'_c)^{0.5} \rightarrow \Delta A_c^{(b)} = \Delta(h^{(c)} d^{(c)}) \cong (\dfrac{f_{yv} A_{sv}^{(c)}}{2hs} + \beta (f'_c)^{0.5})^{-1}\,\Delta V_u^{(c)} = K_{14}\,\Delta V_u^{(c)} \end{cases}$$

(38)

The variation of the perimeter of a square column section, which determines the variation of the column formwork, affects the section capacity as follows:

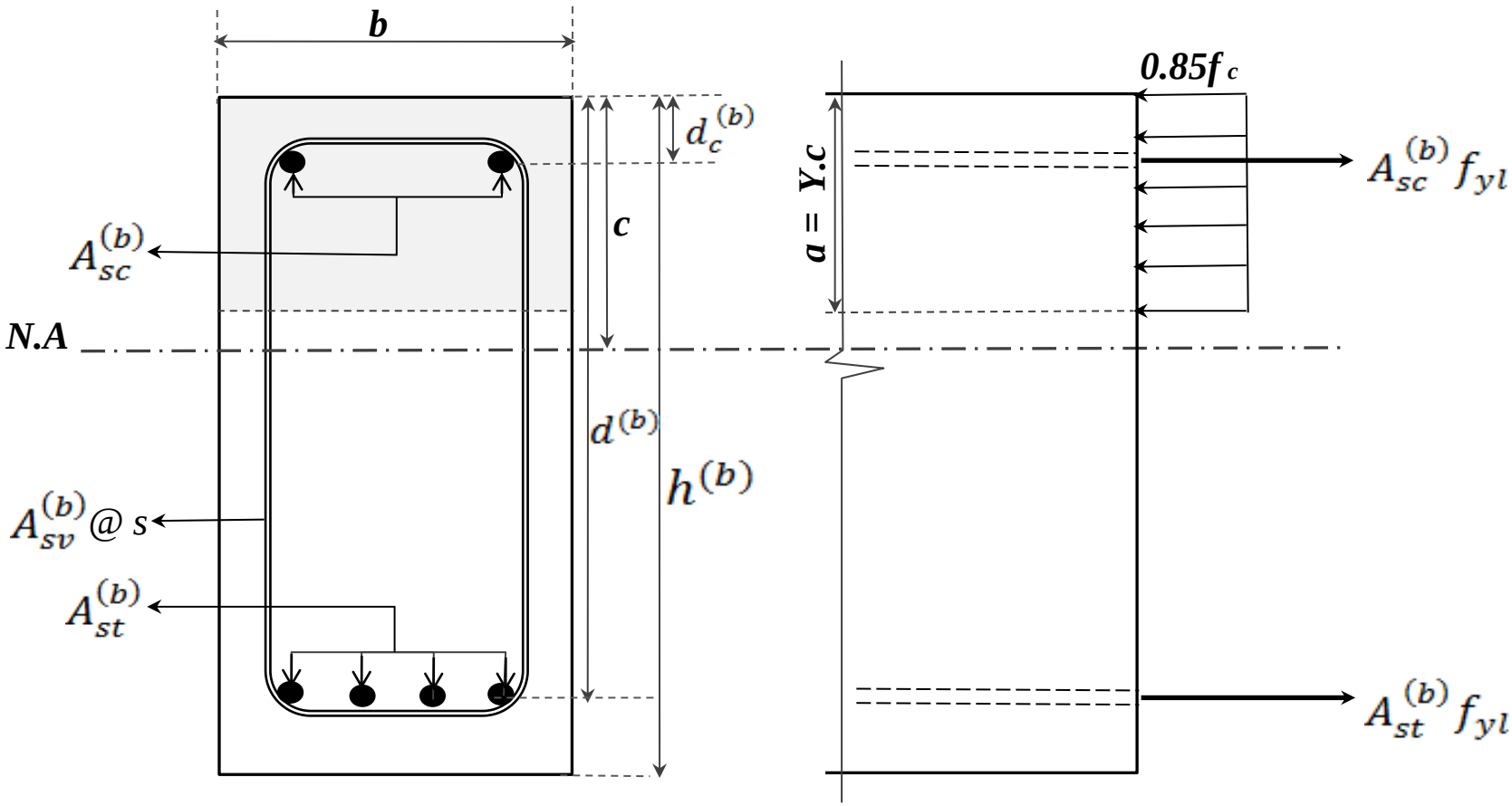
$$\begin{cases} \dfrac{\Delta N_u^{(c)}}{\Delta P_f^{(c)}} \cong 0.85\gamma k_u f'_c \dfrac{h^{(c)}}{2} \rightarrow \Delta P_f^{(c)} \cong (0.85\gamma k_u f'_c \dfrac{h^{(c)}}{2})^{-1}\,\Delta N_u^{(c)} = K_{15}\,\Delta N_u^{(c)} \\ \dfrac{\Delta M_u^{(c)}}{\Delta P_f^{(c)}} \cong 0.85\gamma k_u f'_c \dfrac{h^{(c)}}{2} e_c \rightarrow \Delta P_f^{(b)} \cong (0.85\gamma k_u f'_c \dfrac{h^{(c)}}{2} e_c)^{-1}\,\Delta M_u^{(c)} = K_{16}\,\Delta M_u^{(c)} \\ \dfrac{\Delta V_u^{(c)}}{\Delta P_f^{(c)}} \cong \dfrac{f_{yv} A_{sv}^{(c)}}{4s} + \dfrac{1}{2}\beta h^{(c)} (f'_c)^{0.5} \rightarrow \Delta P_f^{(c)} \cong (\dfrac{f_{yv} A_{sv}^{(c)}}{4s} + \dfrac{1}{2}\beta h^{(c)} (f'_c)^{0.5})^{-1}\,\Delta V_u^{(c)} = K_{17}\,\Delta V_u^{(c)} \end{cases} \quad (39)$$

The reciprocal relationships between the variations of cross-sectional variables and those of strength capacity variables of a column section are obtained by multiplying both sides of Eqs. (9.A) by $c_{sl}/2$, Eqs. (10.A) by c_{sv}, Eqs. (11.A) by $c_c/3$ and Eqs. (12.A) by $c_f/3$ then adding them up :

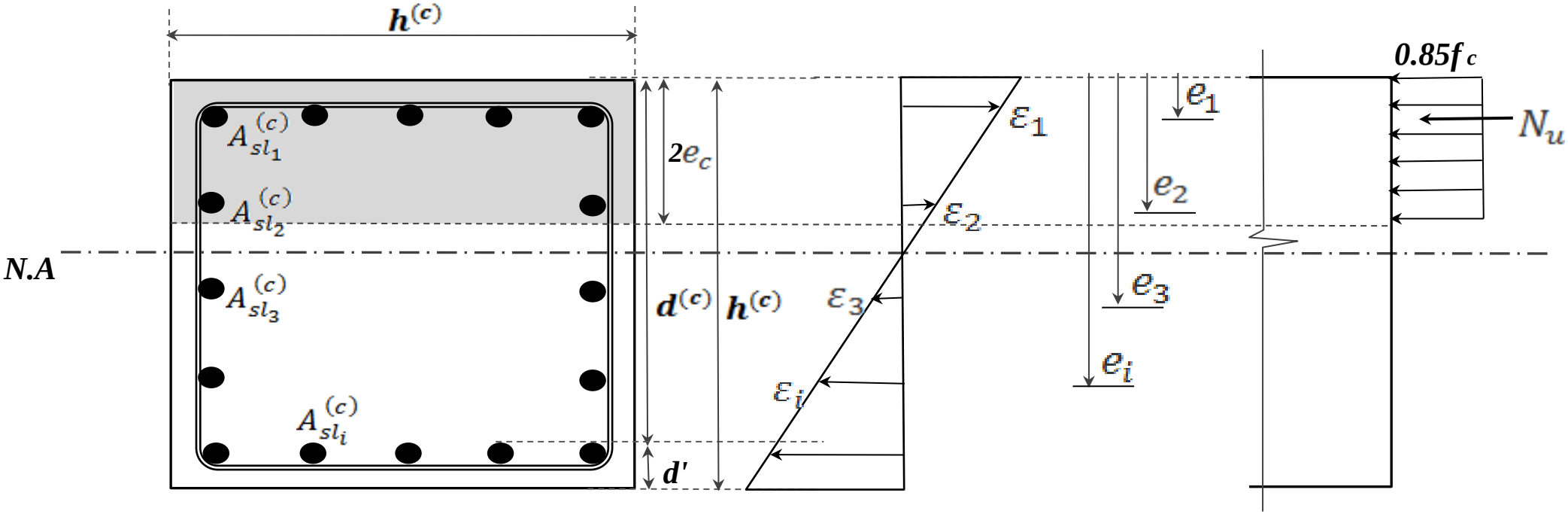
$$\begin{aligned} c_c\,\Delta A_c^{(c)} + c_{sl}\,\Delta A_{sl}^{(c)} + c_{sv}\,A_{sv}^{(c)} + c_f\,\Delta P_f^{(c)} = & \left(\frac{1}{3}c_c K_{12} + \frac{1}{2}c_{sl} K_9 + \frac{1}{3}c_f K_{15}\right)\Delta N_u^{(c)} \\ & + \left(\frac{1}{3}c_c K_{13} + \frac{1}{2}c_{sl} K_{10} + \frac{2}{3}c_f K_{16}\right)\Delta M_u^{(c)} \\ & + \left(\frac{1}{3}c_c K_{14} + c_{sv} K_{11} + \frac{2}{3}c_f K_{17}\right)\Delta V_u^{(c)} \end{aligned}$$

(40)

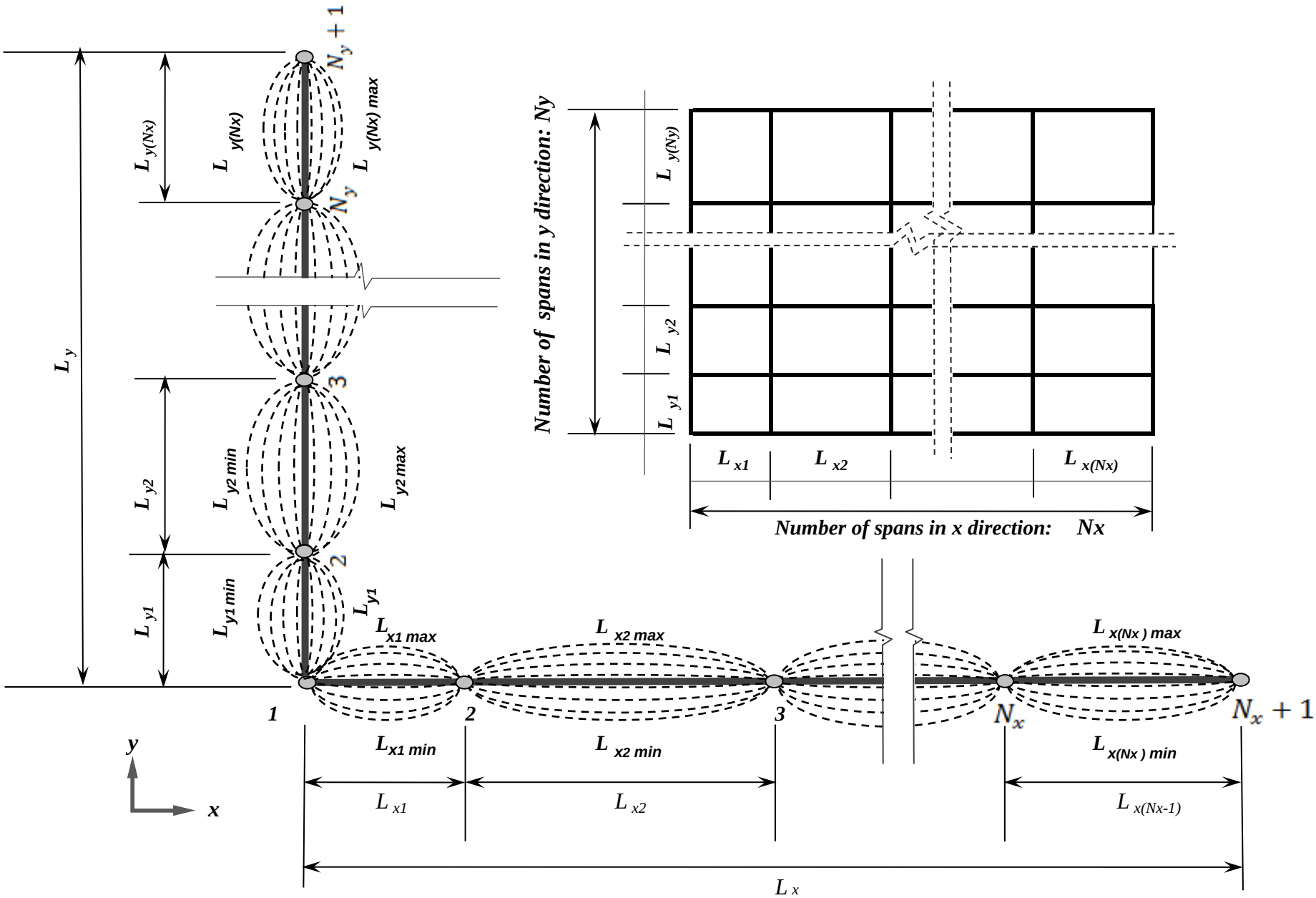
Comparing Eq. (40) with Eqs. (15) and (16) results in Eq. (20).



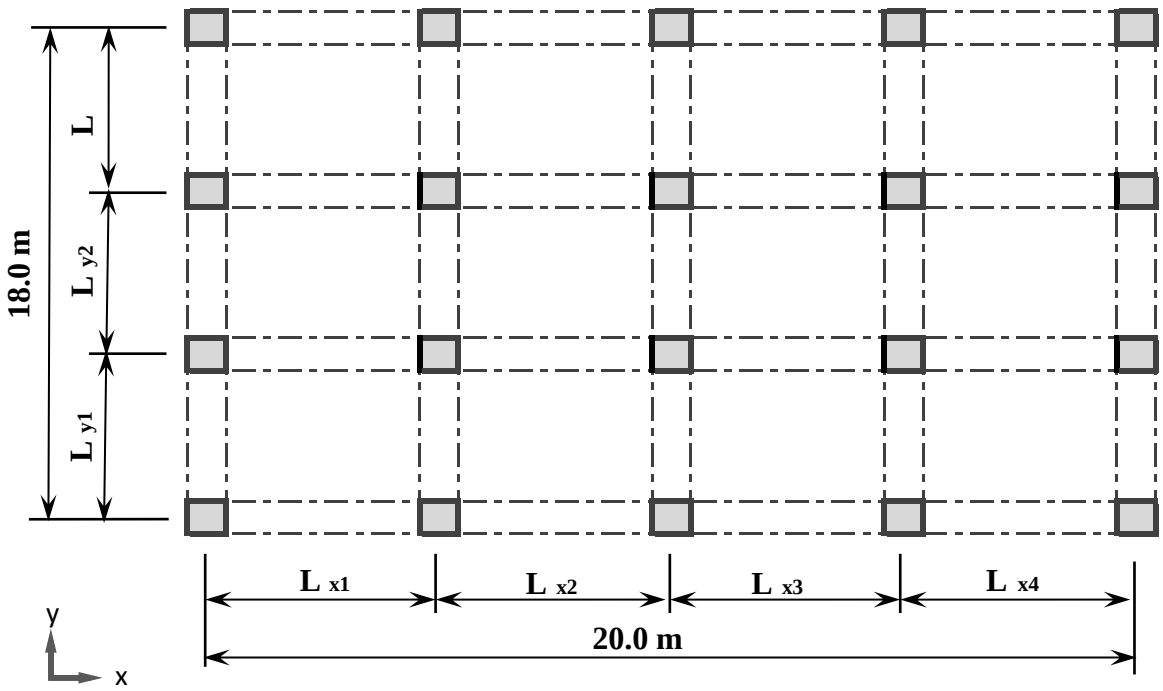
Cross - Section *Equivalent rectangular stress distribution*



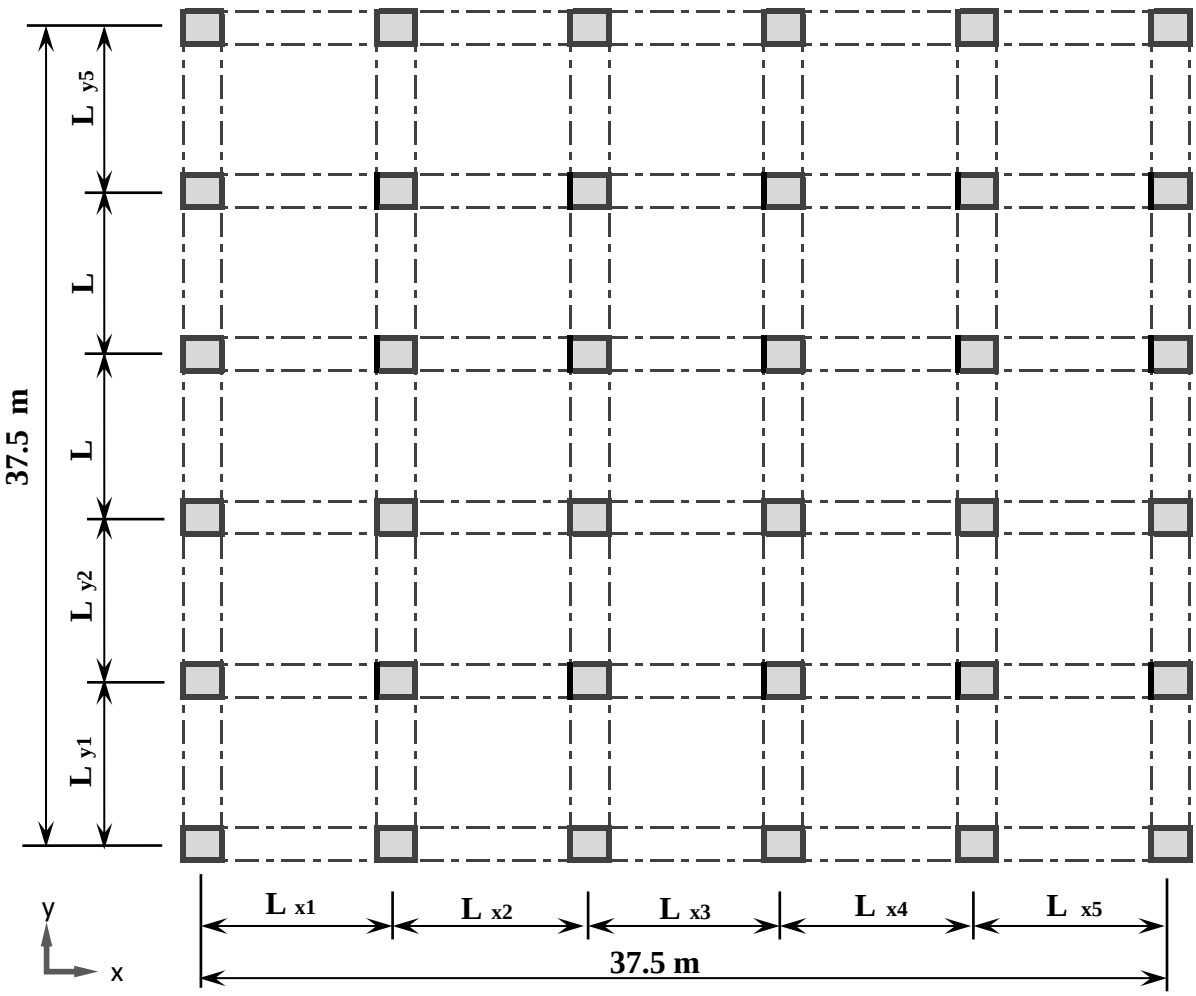
Cross - Section


$N_y + 1$
$L_{y(Nx)}$
$L_{y(Nx)}$
$L_{y(Nx)\ max}$
N_y
L_y
3
L_{y2}
$L_{y2\ min}$
$L_{y2\ max}$
2
L_{y1}
$L_{y1\ min}$
L_{y1}
Number of spans in y direction: Ny
$L_{y(Ny)}$
L_{y2}
L_{y1}
L_{x1}
L_{x2}
$L_{x(Nx)}$
Number of spans in x direction: Nx
$L_{x1\ max}$
$L_{x2\ max}$
$L_{x(Nx)\ max}$
1
2
3
N_x
$N_x + 1$
y
x
$L_{x1\ min}$
$L_{x2\ min}$
$L_{x(Nx)\ min}$
L_{x1}
L_{x2}
$L_{x(Nx-1)}$
L_x

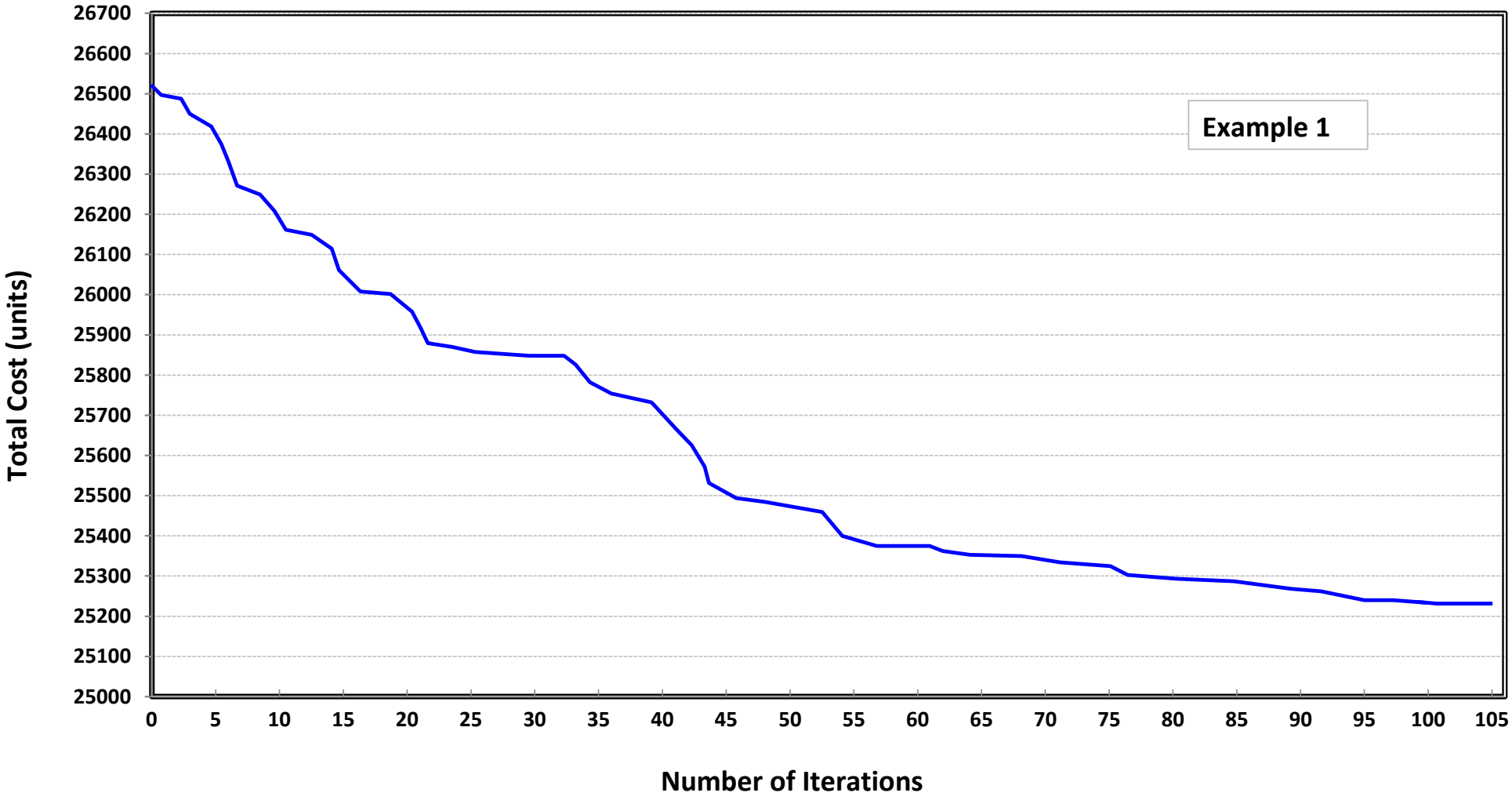
`



(a) Example 1



(b) Example 2


Example 1
Total Cost (units)
26700
26600
26500
26400
26300
26200
26100
26000
25900
25800
25700
25600
25500
25400
25300
25200
25100
25000
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
105
Number of Iterations

(a) Example 1

Example 2
Total Cost (units)
96000
95800
95600
95400
95200
95000
94800
94600
94400
94200
94000
93800
93600
93400
93200
93000
92800
92600
92400
92200
92000
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
105
110
115
120
125
Number of Iterations

(b) Example 2